RAI and HAI: a deep technical analysis of PID-controller stablecoins

RAI and HAI represent the most rigorous application of control theory to cryptocurrency monetary policy ever deployed. RAI, launched February 17, 2021, Medium used a PI controller to maintain a floating "managed peg" backed by ETH, Medium +2 while HAI — its multi-collateral successor on Optimism DL News — launched February 20, 2024 with yield-bearing collateral support addressing RAI's core economic limitation. RAI completed Global Settlement on June 13, 2025, X providing a real-world dataset for how these systems behave across their full lifecycle. Both protocols share the GEB framework architecture, Reflexer a modified MakerDAO MCD fork, OpenZeppelin +2 with permissionless keeper functions that require continuous external calls to remain operational. This report covers every major aspect of both systems' mechanics, vulnerabilities, and historical performance.

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1. How the PI controller creates incentives through redemption rate adjustment

RAI uses a **PI (Proportional-Integral) controller** — not a full PID. Medium The derivative term was intentionally set to zero based on BlockScience's analysis, as it introduced destabilizing noise. At launch, only the P-term was active (**$Kp = 7.5 \times 10^{-8}$** per second per unit of absolute price deviation). The I-term with a leaky integrator was added later to eliminate steady-state error.

The core formula is straightforward: `redemption_rate = Kp × (redemption_price - market_price) + Ki × ∫ error(τ) dτ`. When market price exceeds redemption price, the error is negative, producing a **negative redemption rate** that causes the redemption price to decline over time. When market price is below redemption, the error is positive, producing a **positive redemption rate** that makes the redemption price rise. Reflexer The redemption price updates continuously: `redemption_price(t+1) = redemption_price(t) × (1 + redemption_rate × Δt)`.

When market price trades ABOVE redemption price, the negative redemption rate creates two reinforcing incentive channels. First, anyone with ETH can deposit into a Safe, mint RAI valued at the lower redemption price, and sell it at the higher market price for immediate profit. The **135% minimum collateralization ratio establishes a hard ceiling** — as Dankrad Feist noted, RAI literally cannot trade more than ~35% above redemption for long because the arbitrage becomes risk-free at that point. Dankrad Feist Second, existing Safe holders are incentivized to mint additional RAI because the negative rate means their debt shrinks in real terms over time. For RAI holders, the negative rate creates a "cost of holding" that encourages selling. All three forces push market price downward toward redemption.

When market price trades BELOW redemption price, the positive redemption rate works through a different set of mechanisms. **Safe/vault owners are the primary incentivized party** — their debt is growing in real terms (each RAI owed is "worth more" to the protocol), their borrowing power decreases, and they face escalating liquidation risk. The rational response is to buy RAI cheaply at market and repay their debt, capturing the spread between market and redemption price. Reflexer Reflexer This creates direct buy pressure.

Critically, **non-debt-holders also have meaningful incentives when market price is below redemption**. They can speculate on price convergence, knowing the rising redemption price creates a tailwind for market price appreciation. RAI's documentation explicitly highlights yield stacking: "If Bob is lending RAI at 4%/year and the redemption rate is +10%/year, his net return is **14% annually**." (Reflexer) Additionally, during Global Settlement, each RAI can be redeemed for ETH at the redemption price regardless of market price, giving every RAI holder a fundamental value floor. (Reflexer) Dankrad Feist's analysis demonstrates that a sustained 10% deviation below redemption would cause the integral term to accumulate to ~240% annualized redemption rate, making it economically impossible for Safe holders to maintain positions — forcing massive buy pressure. (Dankrad Feist)

Direct redemption without a vault is impossible during normal operation

A crucial design constraint: **there is no on-demand redemption mechanism** for converting RAI → ETH through the protocol without owning a Safe. RAI explicitly follows MakerDAO's original DAI model rather than Liquity's LUSD direct redemption approach. To burn RAI and recover collateral, you must (1) own a Safe with outstanding debt, (2) repay the RAI debt, and (3) withdraw your ETH. (CoinGecko) The only exception is **Global Settlement** — an emergency shutdown where any RAI holder can redeem at the redemption price. (Albaron Ventures) At RAI's settlement, this rate was fixed at **0.0011267 ETH per RAI**. (X) This distinction matters enormously for low-liquidity scenarios: there is no mechanism for non-vault-holders to arbitrage a discount to redemption in normal operation.

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2. Low liquidity, death spirals, and RAI's orderly wind-down

RAI's official documentation specifies minimum liquidity thresholds for safe PID operation: at least **$2M worth of liquidity** on oracle-feeding exchanges, and at least **3% of RAI supply** must reside on those exchanges. (Reflexer) Below these thresholds, "governance is advised to pause the PID and restart it only after the liquidity improves." (reflexer) (Reflexer) The documentation explicitly cites Proto RAI (PRAI), an unaudited mainnet demo, (Medium) where low Uniswap V2 liquidity enabled market manipulation that destabilized the controller. (reflexer +2)

RAI's actual liquidity trajectory tells a stark story. From a peak of **$300M+ TVL and $100M+ market cap** in April 2021, (CoinGecko) the protocol declined to ~$8M market cap by January 2023 — a 90%+ collapse in usage. (The Defiant) Daily active wallets hit all-time lows by mid-2022. (Albaron Ventures) RAI supply contracted from ~25M to ~560K. (CoinGecko) By early 2026, market cap sat at roughly **$1.7M** (Coinbase) with ~$19K daily trading volume on Uniswap V2. (CoinGecko)

Despite this dramatic decline, **RAI never experienced a death spiral**. The market price stayed within a ~10% band around redemption throughout 2022 ($2.81–$3.10), (The Defiant) with the largest divergence occurring during June 2022's forced liquidations following the Terra/Luna collapse. That divergence closed within weeks. (Staking Rewards) The fundamental reason is Vitalik Buterin's key insight: "The lenders are holding a leveraged position in ETH, not FLX, so there's no risk of a positive-feedback loop where reduced confidence in RAI causes demand for lending to also decrease." (Vitalik Buterin) Unlike UST/Terra, where the backing asset (LUNA)

was endogenous to the system, RAI's ETH collateral exists independently — its value doesn't depend on confidence in RAI. Reflexer

Vitalik's extreme-case analysis demonstrates the ultimate resilience: "If all demand for holding RAI disappears simultaneously except for one holder, the redemption rate would skyrocket until eventually every lender's safe gets liquidated. The single remaining holder would be able to buy the safe in the liquidation auction, use their RAI to immediately clear its debt, and withdraw the ETH." Vitalik Buterin There is always a clean resolution path. The system-wide collateralization ratio remained **above 350%** even during major crashes, providing massive buffer. Bluechip

RAI's wind-down proceeded as follows: In November 2024, Reflexer announced Global Settlement for end of Q1 2025. X On June 13, 2025 (delayed from the original timeline), shutdown was triggered. The oracle freeze activated, Safe owners could claim surplus collateral, and after a 4-day delay, RAI became redeemable at the fixed rate. X Albaron Ventures A notable issue emerged: DeFi Wonderland identified ~~500K RAI (~~$1.5M) locked in the Surplus Buffer that would have been permanently lost during settlement. The Reflexer team was already aware and addressed it. X The wind-down was orderly despite years of declining liquidity — a vindication of the overcollateralized, external-asset-backed design.

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3. The keeper system that keeps everything running

RAI's GEB framework depends on a network of permissionless GitHub "poke" functions that **anyone can call** but that require economic incentives to ensure reliability. If these functions stop being called, the entire system degrades. The protocol runs **15+ distinct pinger bots** deployed as AWS Lambda functions using the Serverless Framework, GitHub each with its own derived Ethereum address from a BIP39 mnemonic.

The critical update cycle follows this sequence:

1. **Medianizer** aggregates oracle feeds (Chainlink or governance-led) into a canonical price
2. **FSM/OSM** (updateResult()) pulls the medianizer price into a delayed queue (~1 hour delay as anti-manipulation defense)
3. **OracleRelayer** (updateCollateralPrice(bytes32)) reads the FSM, internally recalculates redemptionPrice based on elapsed time and current redemptionRate, then computes safetyPrice and liquidationPrice and pushes them to SAFEEngine reflexer
4. **CoinTWAP oracle** updates (~every 12 hours) to provide the market price feed
5. **PIRateSetter** (updateRate()) reads the TWAP market price and current redemption price, computes the error, passes it to the PI calculator, and forwards the new redemptionRate to OracleRelayer via SetterRelayer GitHub github (~every 12 hours)
6. **TaxCollector** (taxSingle(bytes32)) collects accrued stability fees and distributes to AccountingEngine and StabilityFeeTreasury
7. **AccountingEngine** (settleDebt()) cancels matching surplus and debt GitHub

8. **StabilityFeeTreasury** (transferSurplusFunds()) rebalances reserves GitHub

Additional automated parameter bots handle debt ceiling adjustments, auction throttling, debt floor adjustment (via gas oracle), surplus buffer sizing, and system liveness checking. GitHub GitHub

Treasury-funded keeper rewards prevent tragedy of the commons

Keepers are compensated through the **StabilityFeeTreasury**, which receives a portion of the 2% stability fee. Reflexer Rewards accrue as internal SAFEEngine coin balance (not ERC20), requiring a two-step withdrawal process through CoinJoin. The reward structure uses increasing incentives: baseUpdateCallerReward provides a minimum per call, with rewards scaling up to maxUpdateCallerReward based on time elapsed since the last update. This design ensures that even if regular keepers go offline, the increasing reward eventually makes it profitable for someone to step in.

The treasury targets roughly **6 months of operational expenses** as reserve. Reflexer Albaron Ventures At RAI's peak TVL, the 2% stability fee on $50M+ generated meaningful revenue — the protocol spent approximately **246,695 RAI** in total operational costs. Albaron Ventures On Ethereum L1, individual poke transactions cost $5–$50+ depending on gas prices, making keeper economics tight during high-gas periods.

If nobody calls these functions, consequences escalate from stale prices (hours) to broken liquidations and PID malfunction (1–3 days) to potential unbacked debt creation and system failure (days to weeks). The documentation warns of "tragedy of the commons" risk and miner collusion as threats. reflexer No major documented incidents of delayed updates causing real losses were found, suggesting the incentive structure worked adequately throughout RAI's lifetime — though the declining TVL made keeper economics increasingly marginal.

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4. Collateral types and how yield-bearing assets change everything

RAI accepted only ETH Medium with a **145% minimum collateralization ratio**, Albaron Ventures a 2% fixed stability fee, Staking Rewards Albaron Ventures and a 12% liquidation penalty. Albaron Ventures System-wide CR consistently exceeded 350%. Bluechip The minimum debt per Safe was initially high (800+ RAI) due to L1 gas costs needed to make liquidation economically viable for keepers. Albaron Ventures

HAI's multi-collateral design fundamentally changes the economic equilibrium. Medium It accepts **WETH, wstETH, OP, rETH, tBTC, and haiVELO**, TokenInsight each with independent parameters. Mirror The haiVELO collateral type requires **200% minimum collateralization** given its volatility, with a 10% performance fee on veVELO rewards set aside as a bad-debt backstop. More volatile assets (OP, VELO) require higher collateralization ratios than stable assets (WETH, wstETH).

Yield-bearing collateral addresses RAI's core economic flaw. RAI suffered persistent negative redemption rates because Safe owners faced opportunity cost: locking ETH instead of staking it for ~3–5% APY. Medium Mirror As co-founder Ameen Soleimani admitted in January 2023, "ETH-only RAI was a mistake." The Defiant

With wstETH/rETH as collateral, minters earn staking yield while simultaneously borrowing HAI — eliminating the opportunity cost. (Mirror) Soleimani predicted "equilibrium redemption rates for HAI will probably trend positive, so the price of HAI will get higher over time." (reflexer)

The PID controller is **collateral-agnostic** — it only measures market price versus redemption price deviation. Yield-bearing collateral affects the controller indirectly: wstETH's appreciating value naturally improves vault collateralization ratios over time (collateral grows while debt stays constant), changing minter incentives and therefore supply/demand dynamics. Volatile collateral can stress the PID through mass liquidation cascades that rapidly alter HAI supply, but independent debt ceilings per collateral type limit systemic contagion from any single volatile asset.

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5. Battle testing: RAI survived every major ETH crash

May 2021 (ETH -50%, ~$4,300 → $1,800): RAI's first major stress test at just 3 months old. Cointelegraph reported RAI "hovered close to its algorithmically generated target price of $3 throughout the downturn." (Cointelegraph) With $300M+ TVL providing deep liquidity, the PID controller adjusted smoothly.

June–July 2022 (ETH -50%, ~$1,800 → $880, post-Terra/Luna): RAI's price dropped from ~$3.00 to the ~$2.85 range. (Bitcoin Insider) (Medium) The divergence between market and redemption price widened after forced liquidations but "has since closed." (Staking Rewards) MixBytes' security analysis concluded: "The algorithm proved to be very adaptive even to the bearish market: currently there are no unhealthy positions to liquidate." (MixBytes) Notably, Dune Analytics data showed whales buying RAI as a safe haven during the 3AC bankruptcy liquidation — an unexpected use case. (Bitcoin Insider)

November 2022 (FTX collapse, ETH below $1,100): No catastrophic RAI event documented. The system continued functioning normally, though by this point liquidity was already severely diminished (~$8M market cap). The stress test was less meaningful because RAI was a small, declining protocol.

HAI since February 2024: No major depegging events have been publicly reported. As a "float coin," small deviations from $1 are by design rather than failures. Specific TVL and volume data were difficult to obtain, but HAI was reportedly the "largest CDP protocol on Optimism" with "over 45% of TVL across all CDPs on OP." (Medium) KITE staking launched in May 2025 with protocol surplus streaming to stakers.

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6. The Moby Dick attack and why the leaky integrator matters

BlockScience named this attack vector in their April 2021 collaboration with Reflexer. The scenario: **a malicious whale accumulates a large RAI position and forcibly pins the market price at a constant value**, preventing convergence to the redemption price. (Medium) This creates a sustained, non-zero error signal that the controller must respond to.

With a pure P controller, the whale can maintain the offset indefinitely — proportional control cannot eliminate steady-state error. With a standard PI controller (no leak), the integral term accumulates this persistent error unboundedly, causing **integral windup**. Even after the whale exits, the accumulated integral creates massive overshoot — the redemption price has drifted so far that the system becomes destabilized.

The **leaky integrator** solves this by introducing exponential decay on the integral term: `I(t) = ∫ e(τ) · e^(-α(t-τ)) dτ`, where α is the leak rate. Older errors are progressively "forgotten." This means the integral **reaches a bounded maximum** regardless of how long the whale maintains the attack. Once the whale releases their position, recovery is rapid because there's minimal accumulated bias to unwind. BlockScience's simulations confirmed: "Non-leaky PI controllers generate a drift on the redemption price even though the magnitude of the cumulative price error remains bounded" (Medium) — the leaky integrator eliminates this drift.

The leak rate α is a critical governance parameter: too high and the integral loses its ability to correct steady-state error; too low and it approaches the vulnerable non-leaky behavior. BlockScience used cadCAD modeling with stochastic ETH price processes to identify safe parameter ranges through their "Parameter Selection under Uncertainty" methodology. (Medium) (GitHub)

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7. Smart contract architecture and the full GEB update flow

The GEB framework centers on **SAFEEngine** (`0xCC88a9d330da1133df3A7bD823B95e52511A6962` on mainnet) — the self-contained central accounting ledger storing all Safe states, coin balances, collateral amounts, and debt. It has no external dependencies. Key surrounding contracts include:

- **OracleRelayer** (`0xe4e834fc6808181668501E89a85F1cC40ad941d9`): The critical glue between oracle feeds and SAFEEngine. Stores `redemptionPrice` and `redemptionRate`. (reflexer) (Reflexer) When `updateCollateralPrice()` is called, it reads the FSM price, recalculates the redemption price based on elapsed time (`rpow(redemptionRate, elapsed, RAY) × redemptionPrice`), computes `safetyPrice = (feedPrice × 10⁹) / redemptionPrice / safetyCRatio` and `liquidationPrice` similarly, then pushes both to SAFEEngine. (reflexer)
- **PIRateSetter** (`0x7Acfc14dBF2decD1c9213Db32AE7784626daEb48`): Reads TWAP market price and redemption price, computes error, feeds to PI calculator, and passes the resulting rate through **SetterRelayer** to OracleRelayer. (github)
- **TaxCollector**: Collects per-collateral stability fees via `taxSingle()`, distributing to AccountingEngine (primary) and StabilityFeeTreasury (secondary). (OpenZeppelin)
- **LiquidationEngine**: Checks if (reflexer) `lockedCollateral × liquidationPrice < generatedDebt × accumulatedRate`, triggering collateral auctions via FixedDiscountCollateralAuctionHouse.
- **FSM/OSM**: Introduces ~1-hour price delay from medianizer feeds as an anti-flash-manipulation defense.
- **StabilityFeeTreasury**: Funds keeper operations with per-address withdrawal allowances. (Reflexer)

HAI uses the identical architecture with extensions for multi-collateral support: Medium Letsgethai multiple collateral join adapters, independent oracle feeds per collateral type, per-collateral parameters in OracleRelayer and TaxCollector, and the addition of a **Stability Pool** for automated liquidation participation Letsgethai (not present in original RAI).

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8. Audit findings and known vulnerability surface

The **OpenZeppelin audit** (November 13, 2020) of core GEB contracts found **zero critical issues and one high-severity issue**. OpenZeppelin openzeppelin Notable findings included: English auction vulnerability to network congestion (mitigated by switching to fixed-discount auctions), openzeppelin a totalAuctionLength initialization concern (acknowledged as intentional), fee collection edge cases in the CoinSavingsAccount, and risks from malicious Saviour contracts that could affect system parameters if governance whitelists them carelessly. OpenZeppelin openzeppelin All issues were addressed. OpenZeppelin The audit explicitly excluded governance mechanisms, oracle price feeds, and tests from scope. OpenZeppelin openzeppelin

RAI also received audits from **Quantstamp** (two engagements covering helper contracts and the geb-lender-first-resort module) and **Solidified**. reflexer All three firms' reports are archived in the reflexer-labs/geb-audits GitHub repository. GitHub

HAI maintains an audit-reports repository at github.com/hai-on-op/audit-reports, GitHub and DeFi Llama confirms audits were conducted. DefiLlama However, specific auditor names, severity findings, and scope details were not publicly accessible during this research. Being a GEB fork Medium developed by DeFi Wonderland Medium means HAI inherits the audited codebase foundation, but modifications for multi-collateral support, Optimism deployment, and novel features like haiVELO introduce fresh attack surface requiring independent review.

No exploits or hacks of either RAI or HAI have been documented. The primary ongoing vulnerability surfaces are: oracle manipulation during low-liquidity conditions Elliptic reflexer (mitigated by TWAP but not eliminated), Medium PID controller parameter risks (too aggressive destabilizes, too passive is ineffective), reflexer +4 Saviour contract risks from malicious whitelisting, reflexer and the fundamental dependency on continuous keeper operation and external oracle infrastructure reflexer — areas where governance minimization is inherently limited.

Conclusion

RAI and HAI validated that PID controllers can stabilize floating-peg stablecoins through multi-year operation including severe market crashes Albaron Ventures — but also revealed fundamental limitations. RAI's single-ETH-collateral model created persistent negative redemption rates from staking opportunity cost, Medium The Defiant ultimately contributing to declining adoption and the "ETH-only was a mistake" admission. HAI's yield-bearing collateral directly addresses this by allowing wstETH/rETH minters to earn staking yield while borrowing, Medium potentially enabling positive equilibrium redemption rates. Mirror

The most important architectural insight is the **absence of direct redemption**: non-vault-holders cannot burn RAI/HAI for collateral during normal operation, meaning price convergence depends entirely on indirect incentives (Safe holder arbitrage, speculation, yield seeking) rather than mechanical arbitrage. This makes liquidity depth existentially important — below the documented $2M / 3%-of-supply thresholds, the PID controller's feedback loop can break down. reflexer Reflexer RAI's overcollateralization with external assets (ETH) proved absolutely decisive in preventing death spirals despite 90%+ usage decline, Reflexer drawing a bright line between systems like RAI and endogenous-collateral designs like UST that collapsed catastrophically. BlockApps Inc. Vitalik Buterin The keeper infrastructure, while functional, represents a persistent centralization risk: 15+ AWS Lambda functions with treasury-funded rewards are the system's heartbeat, GitHub and their failure would render the protocol non-functional within days.